

CO1 : Technical Presentation.

CSA1310 : Theory of Computation.

1. NFA to DFA Conversion

$$\Sigma = \{0, 1\}$$

States:

$$Q = \{A, B, C\}$$

Start state: A

Final state: C

State	Input	Output
A	{A}	{B}
B	{A, C}	$\emptyset$
C	$\emptyset$	{A, B}

Step: 1

Start state: {A}

For Input

$$(\{A\}, 0) = \{A\}$$

For Input: 1

$$(\{A\}, 1) = \{B\}$$

So, DFA:

$$\{A\} \{B\}$$

Step: 2

For Input: 0

$$\{B\}, \{0\} = \{A, C\}$$

For Input: 1

$$(\{B\}, 1) = \emptyset$$

New states:

$$\{A, C\}, \emptyset$$

Step 3 Process $\{A, c\}$

$A \rightarrow A$
 $c \rightarrow \emptyset$

Therefore:

$$(\{A, c\}, 0) = \{A\}$$

For Input 1:

$A \rightarrow B$
 $c \rightarrow \{A, B\}$

$$\text{Therefore: } (\{A, c\}, 1) = \{A, B\}$$

New State: $\{A, B\}$

A.C. is a final State.

~~is a final State~~

Step 4

process $\{A, B\}$

For Input 0:

$A \rightarrow A$
 $B \rightarrow \{A, c\}$

Therefore:

$$(\{A, B\}, 0) = \{A, c\}$$

Step 5 process $\emptyset$

For both inputs

$$(\emptyset, 0) = \emptyset$$

$$(\emptyset, 1) = \emptyset$$

$\emptyset$ is the dead state.

Final Transition Table (DFA)

DFA State	0	1	Final	Start	Final
$\rightarrow \{A\}$	$\{A\}$	$\{B\}$	NO		
$\{B\}$	$\{A, c\}$	$\emptyset$	NO		
$\star \{A, c\}$	$\{A\}$	$\{A, B\}$	YES		
$\{A, B\}$	$\{A, c\}$	$\{B\}$	NO		
$\emptyset$	$\emptyset$	$\emptyset$	NO		

Mathematical Introduction is a proof technique used to that a statement is true for all positive integers

- strings of length n
- Languages
- Number of symbols in a string
- Properties of automata after reading n symbols

Step: 1

Best case

First, Prove that the statement is true for the value $n=1$

we, Prove $\Rightarrow P(1)$ is true

Step: 2

Inductive hypothesis

$P(k)$ is true

Assume that the statement is true for some integer k . This is assumption is called inductive hypothesis

Step: 3

Inductive step.

$$P(k+1)$$

$$P(k) = P(k+1)$$

using the assumption that $P(k)$ is true prove that $P(k+1)$.

1. $P(1)$ is true
2. $P(k) = P(k+1)$

So,
 $P(n)$ is true for all positive integers n .

Assume:-

$$1^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

For $k=1$

$$\Rightarrow \frac{1(1+1)(2 \cdot 1 + 1)}{6} = 1 + (1+1)^2$$

$$\Rightarrow \frac{(k+1)(k+2)(2k+3)}{6}$$

So,

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

States

Alphabet

Transitions

FROM

C

SYMBOL

1

TO

B

Add transition

A → 0 → A

A → 1 → B

B → 0 → A

B → 0 → C

C → 1 → A

C → 1 → B

Delete transition

DFA: binary strings ending in 1

NFA

Properties

Convert

NFA → DFA (subset construction)

Minimize DFA

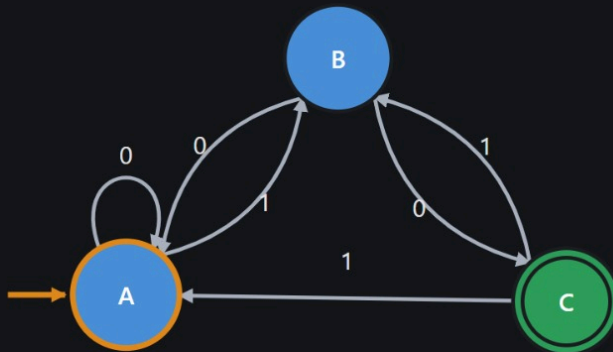
DFA → NFA

SUBSET CONSTRUCTION TABLE

DFA state	NFA states	0	1
→ {A}	{A}	{A}	{B}
{B}	{B}	{A, C}	∅
* {A, C}	{A, C}	{A}	{A, B}
{A, B}	{A, B}	{A, C}	{B}

→ start state · * accepting state · 4 DFA states from 3 NFA states.

Use as current automaton



Subset construction complete: 4 DFA states.



State



Accepting



Start



Dead



Unreachable



Active

INPUT STRING

e.g. 1011

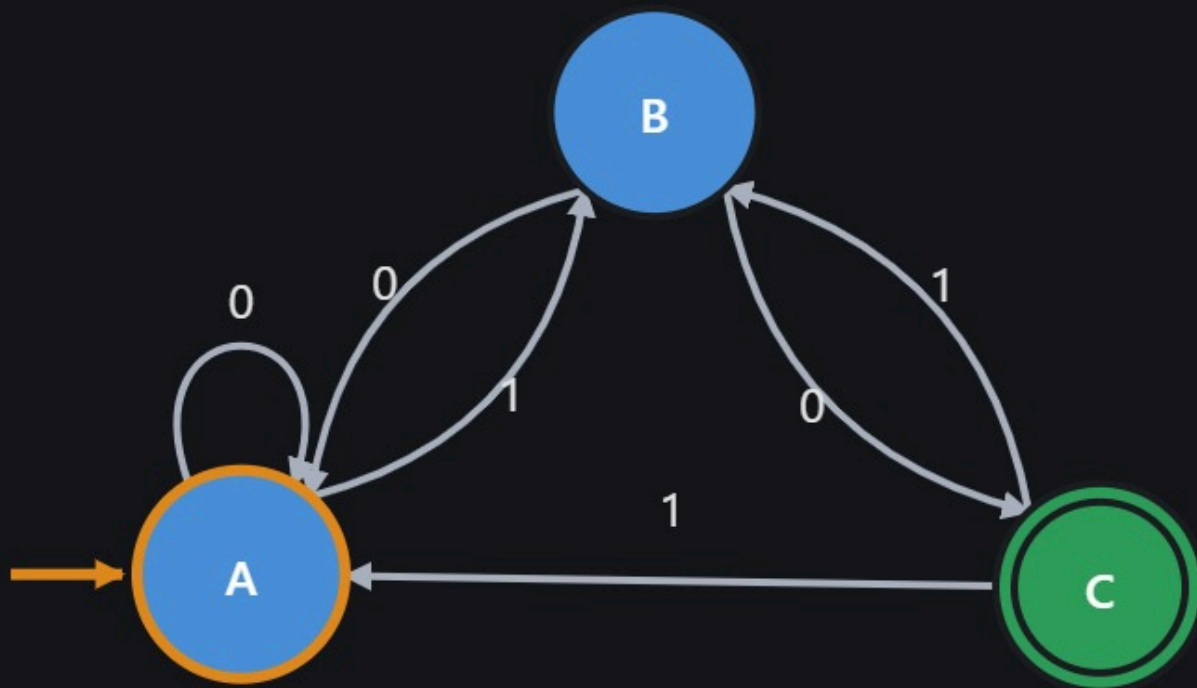
Simulate



Auto



Enter a string and press Simulate.



States

Alphabet

Transitions

FROM

A

SYMBOL

0

TO

C

Add transition

A $\xrightarrow{0}$ BA $\xrightarrow{0}$ CA $\xrightarrow{1}$ AA $\xrightarrow{\epsilon}$ BB $\xrightarrow{1}$ BB $\xrightarrow{\epsilon}$ CC $\xrightarrow{0}$ CC $\xrightarrow{1}$ C

Delete transition

DFA: binary strings ending in 1

NFA

Properties

Convert

NFA $\rightarrow$ DFA (subset construction)

Minimize DFA

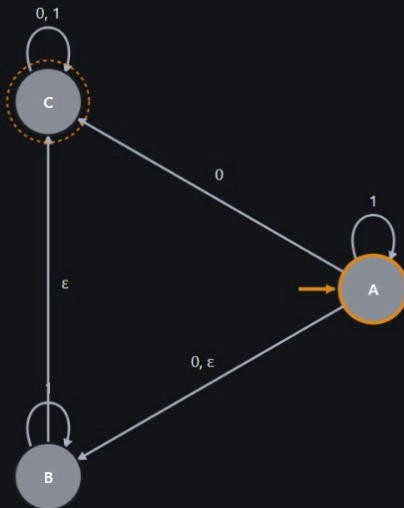
DFA $\rightarrow$ NFA

SUBSET CONSTRUCTION TABLE

DFA state	NFA states	0	1
$\rightarrow$ {A,B,C}	{A, B, C}	{B,C}	{A,B,C}
{B,C}	{B, C}	{C}	{B,C}
{C}	{C}	{C}	{C}

$\rightarrow$ start state · * accepting state · 3 DFA states from 3 NFA states.

Use as current automaton



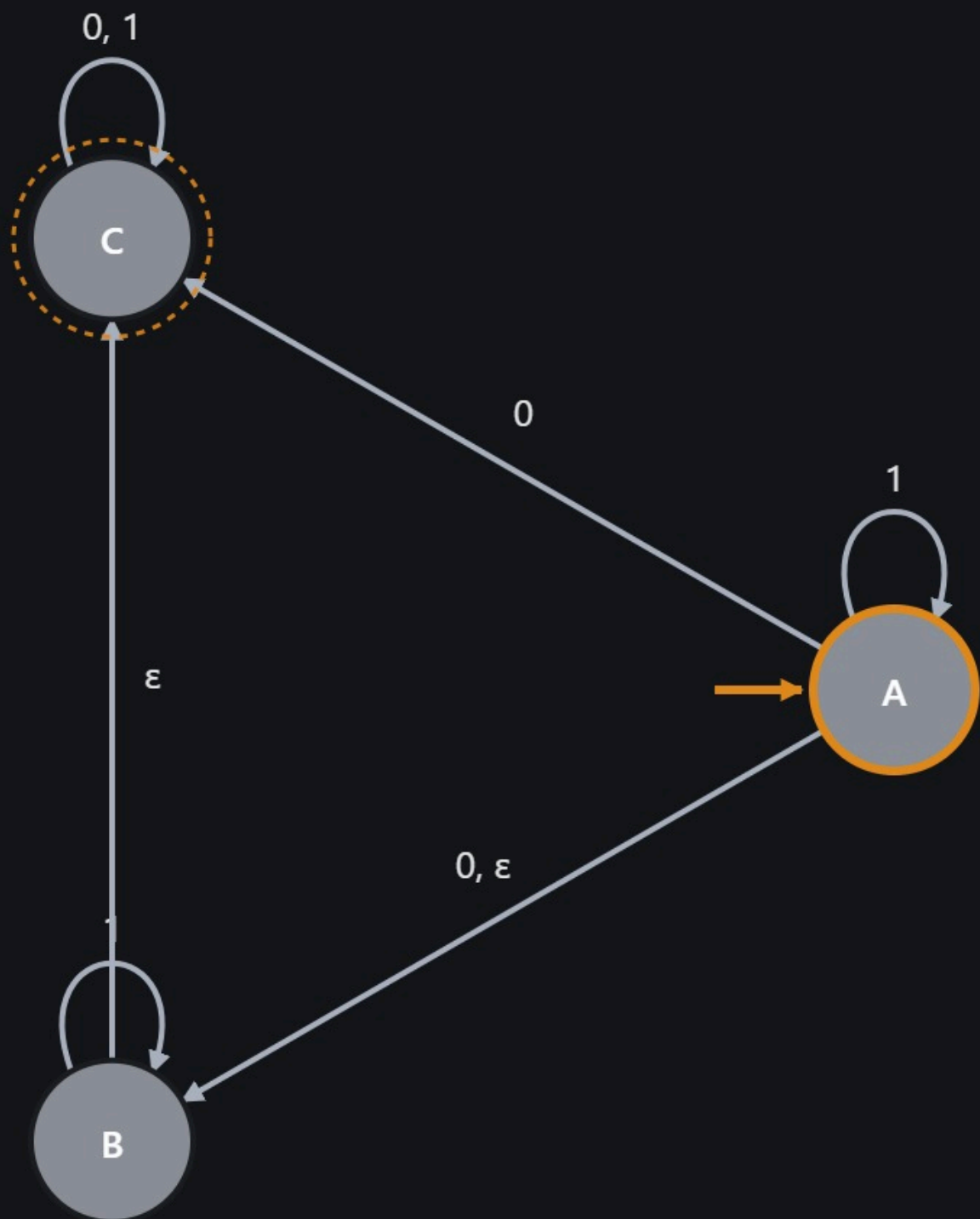
● State ● Accepting ● Start ● Dead ● Unreachable ● Active

INPUT STRING e.g. 1011

Simulate



Enter a string and press Simulate.



States

Alphabet

Transitions

FROM

q1

SYMBOL

 ϵ (epsilon)

TO

q2

Add transition

 $q_0 \xrightarrow{1} q_1$ $q_0 \xrightarrow{\epsilon} q_1$ $q_1 \xrightarrow{0} q_0$ $q_1 \xrightarrow{0} q_1$ $q_1 \xrightarrow{\epsilon} q_2$

Delete transition

DFA: binary strings ending in 1

NFA

Properties

Convert

NFA $\rightarrow$ DFA (subset construction)

Minimize DFA

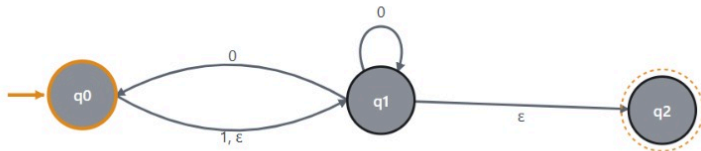
DFA $\rightarrow$ NFA

SUBSET CONSTRUCTION TABLE

DFA state	NFA states	0	1
$\rightarrow \{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$
$\{q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\emptyset$

→ start state · * accepting state · 2 DFA states from 3 NFA states.

Use as current automaton



● State ● Accepting ● Start ● Dead ● Unreachable ● Active

INPUT STRING e.g. 1011

Simulate



▶ Auto



Enter a string and press Simulate.

